

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 2 – Short Answer**

|                  |                                                                                                                       |               |                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------|
| Course Code:     | ITA03                                                                                                                 | Course Title: | Mobile computing                 |
| CO Assessed:     | CO1                                                                                                                   | Assessment:   | Assessment Tool 2 – Short Answer |
| Weightage:       | 30 % of CIA                                                                                                           | Total Marks:  | 100Marks                         |
| CO1 Statement:   | Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. |               |                                  |
| Student Name:    | ROSE MYSTICA-J                                                                                                        | Reg. No.:     | 192421410                        |
| Section / Batch: | D / 3 <sup>rd</sup> year                                                                                              | Date:         | 23.7.26                          |

**Code of Conduct**

I ROSE MYSTICA-J (192421410) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: J. Rosemythia

**Instructions**

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

95  
100

ANALYTICAL QUESTIONS

- 1) Full duplex = 2 simplex channel =  $2 \times 25 = 50 \text{ KHz}$ . ,  $45 \text{ MHz} = 45000 \text{ kHz}$   
 Total spectrum =  $\frac{45000}{50} = 900$ .
- 2) No. of channel a single cell will be allotted } =  $\frac{\text{total voice channels}}{\text{frequency cluster}}$   
 $= \frac{560}{7} = 80$
- 3) Total no. of voice channels = Carriers  $\times$  time slots  
 $= 100 \times 8$   
 $= 800$
- 4) Power ratio =  $\frac{80}{4} = 20 \text{ W}$   
 Power gap in decibel =  $10 \times \log_{10}(20) = 13.02 \text{ dB} = 13 \text{ dB}$ .
- 5) Sign C/I ratio = signal - co-channel interference  
 $= (-60) - (-92)$   
 $= 32 \text{ dB}$ .
- 6) Total channels =  $\frac{42000}{300} = 140$ .  
 channels per cell =  $140 \times 7 = 20$ .
- 7) Received power = doubling distance with  $n=4$ .  
 $2^4 = 16 \times$   
 New power =  $64/16 = 4 \text{ W}$ .

8) Inverse power law  $n=2$ . After doubling the  $n$  we get  $2 \times 2 = 4 \times$ .

9) Total channel count = cluster size  $\times$  voice channels  
 $= 4 \times 15$   
 $= 60$ .

10) Given:  $100 \text{ km/h} = 27.78 \text{ m/s}$ .

$$\text{Doppler shift} = v \times f / c = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8}$$
$$= 83.34 \text{ Hz}$$

11) Slot duration =  $\frac{\text{Frame time}}{\text{Slots}} = \frac{5 \text{ ms}}{8} = 0.625 \text{ ms}$

12) Carriers =  $\frac{\text{Total spectrum}}{\text{spacing}} = \frac{24000 \text{ kHz}}{200 \text{ kHz}} = 120$ .

13) T in power in dBm =  $10 \log_{10}(8000 \text{ mW}) = 39.03 \text{ dBm}$   
Received power =  $39.03 - 33$   
 $= 6.03 \text{ dB}$ .

15) Total capacity =  $\frac{\text{calls}}{\text{calls}} \times \text{cells} = 250 \times 6 = 15000$

16) Active calls supported = arrival time  $\times$  avg duration  
 $= 4000 \times 180 = 720,000$ .

17) Total bandwidth = channels  $\times$  occupying channels.  
 $= 150 \times 25 \text{ kHz} = 3750 \text{ kHz}$   
 $= 3.75 \text{ MHz}$ .

18) Total guard bands between 30 frequency bands = 29.

$$\text{Speed} = 90 \text{ km/h} = 25 \text{ m/s}$$

$$\text{Cell diameter} = 2 \times 1.5 = 3000 \text{ m}$$

$$\text{Time per cell} = \frac{3000}{25} = 120 \text{ s}$$

$$\text{Handoff per hour} = \frac{3600}{120} = 30$$

$$\begin{aligned} \text{Total users} &= \text{regional center} \times \text{base stations / center} \times \text{users / base station} \\ &= 8 \times 40 \times 250 \\ &= 80,000 \end{aligned}$$

$$21) \text{ Approximate area} = R^2 = (0.4)^2 = 0.16 \text{ km}^2$$

$$\text{No. of cells required} = \frac{400}{0.16} = 2500$$

$$22) \text{ No. of channels} = \frac{\text{Total spectrum}}{\text{channel width}}$$

$$= \frac{6000}{30} = 200$$

$$23) \text{ Total call / minutes} = 1200 \times 4 \times 2.5 = 12,000 \text{ min/hr}$$

$$\text{Traffic in Erlangs} = \frac{12000}{60} = 200$$

$$24) \text{ Users} = \frac{\text{Total bandwidth}}{\text{bandwidth per user}} = \frac{40000 \text{ KHz}}{400 \text{ KHz}} = 100$$

$$25) \text{ Total capacity} = \text{user / cell} \times \text{cells} = 600 \times 1200 = 720,000$$

$$26) \text{ Paging Operations} = \frac{\text{Total calls}}{\text{Cells per operation}}$$

$$= \frac{200}{4} = 50$$

$$27) \text{ Offered traffic} = \text{arrival rate} \times \text{avg duration}$$

$$= 40,000 \times 3 = 120,000 \text{ Erlangs}$$

$$\text{Over capacity} = 120,000 - 100,000 = 20,000$$

$$\text{Fraction blocked} = \frac{20,000}{120,000} = 0.17.$$

$$28) \text{ Roaming users} = 15\% \times 600,000$$

$$= 90,000.$$

$$29) \text{ Utilization} = \frac{\text{Peak traffic}}{\text{Processing rate}}$$

$$= \frac{9000}{12000} = 0.75 = 75\%.$$

$$30) \text{ Total channels} = \frac{\text{Bandwidth}}{\text{channel width}} = \frac{48000}{200} = 240.$$

$$\text{channels per cell} = \frac{240}{4} = 60.$$

$$31) \text{ channels} = \frac{\text{Total spectrum}}{\text{channel width}} = \frac{30000}{200} = 150.$$

$$32) \text{ Successful handoffs} = \frac{99}{100} \times 2500 = 2475.$$

~~33)~~

$$34) \text{ Total capacity} = \text{users/cells} \times \text{cells} = 900 \times 7 = 6300.$$

$$\text{slot duration} = \frac{\text{Frame time}}{\text{slots}} = \frac{4.8}{8} = 0.6$$

36)

$$37) \text{ Ordered traffic} = 70,000 \times 4 = 280,000 \text{ Erlangs}$$

$$\text{Excess} = 280,000 - 250,000 = 30,000$$

$$\text{Fraction blocked} = \frac{30,000}{280,000} = 0.11$$

$$38) \text{ Total users in cluster} = \text{users/cell} \times \text{cells} \\ = 600 \times 9 = 5400$$

$$39) \text{ Calls dropped/sec} = 0.4\% \times 60,000 = 240$$

$$40) \text{ Path loss} = T_{\text{power}} - R_{\text{x power}} = 43 - (-87) = 130 \text{ dB}$$

$$41) \text{ Total coverage} = \frac{\text{coverage}}{\text{station}} \times \text{no. of station} \\ = 2.5 \times 150 = 375 \text{ km}^2$$

$$42) \text{ Total channels} = \frac{48000}{200} = 240$$

$$\text{channels per second} = \frac{240}{6} = 40$$

$$43) \text{ Paging cycles} = \frac{\text{Total users}}{\text{users per cycle}} \\ = \frac{150,000}{6000} = 25$$

44)

$$\text{Hand off per cell} = \frac{\text{cell duration}}{\text{handoff interval}} = \frac{9}{3} = 3$$

$$\begin{aligned}
 45) \quad \text{SNR} &= \text{signal} - \text{Noise} \\
 &= -85 - (-115) \\
 &= 30 \text{ dB}
 \end{aligned}$$

$$\begin{aligned}
 46) \quad \text{Minimum bandwidth} &= \text{users} \times \text{rate} / \text{user} \\
 &= 250 \times 150 \text{ kbps} \\
 &= 37.5
 \end{aligned}$$

$$\begin{aligned}
 47) \quad \text{Total calls/hr} &= 15000 \times 4 \times 2.5 \\
 &= 15,000 \text{ min}
 \end{aligned}$$

$$\begin{aligned}
 48) \quad \text{Total bandwidth} &= \text{channels} \times \text{bandwidth/channel} \\
 &= 250 \times 12 \text{ KHz} \\
 &= 3000 \text{ KHz} = 3 \text{ MHz}
 \end{aligned}$$

$$\begin{aligned}
 49) \quad \text{Total throughput} &= \text{connection} \times \text{rate} \\
 &= 1500 \times 60 \text{ kbps} \\
 &= 900,000 \text{ kbps}
 \end{aligned}$$

$$50) \quad \text{Area per tower} = \pi R^2 = \pi \times 4^2 = 3.14 \times 4^2 = 50.27 \text{ km}^2$$

$$\text{Tower needed} = \frac{350}{50} = 7$$